

Optimizing the Falcon PQC Algorithm: Reducing The Execution Time

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Abstract—The quantum computing continues to advance, the security of conventional cryptographic systems faces significant risks, prompting rapid developments in post-quantum cryptography (PQC). This paper presents a series of targeted improvements to the Falcon digital signature scheme in post-quantum cryptography. The analysis of Falcon's framework, several areas can be identified where optimizations could enhance performance. This lowers the computing overhead while maintaining the algorithm's resilience to quantum attacks by using GPU offloading, parameter recalibration, and other optimization strategies. Experimental evaluation shows an approximate 11% reduction in execution time without compromising the security guaranteed by the original method. These enhancements make it more practical for resource-constrained environments.

Index Terms—Post-Quantum Cryptography, Falcon, Lattice-based Signatures, Optimization, GPU Acceleration, FFT, Gaussian Sampling

I. INTRODUCTION

A. The Need for Post-Quantum Cryptography

The advent of quantum computing poses a critical threat to classical cryptographic algorithms such as RSA and ECC [1], which rely on the hardness of integer factorization and discrete logarithm problems. Quantum algorithms like Shor's algorithm [2] undermine the security assumptions of these systems. Thus, post-quantum cryptography (PQC) has emerged to address the challenge, aiming to develop schemes resilient to quantum attacks. Efforts such as NIST's PQC standardization project have pushed multiple algorithms into evaluation, including Falcon [1].

B. The Falcon Algorithm: Promise and Challenges

Falcon [1] (Fast Fourier Lattice-based Compact Signatures over NTRU) [3] is a lattice-based signature scheme selected by NIST [4], designed with compactness and strong theoretical guarantees. Based on the hardness of the Short Integer Solution (SIS) problem over NTRU lattices [5], Falcon is expected to resist both classical and quantum attacks and its performance against classical [6] algorithm. Despite its strengths—such as compact signatures and efficient verification, it suffers from expensive operations similar to FFTs and discrete Gaussian sampling, making performance a key concern for deployment on constrained devices.

C. Why Performance Optimization Matters

Optimization enhances the practicality of PQC in real-world systems:

- **Deployability:** Cryptographic functions are often latency-sensitive.
- **Energy Efficiency:** Especially crucial for mobile and embedded systems.
- **Side-Channel Resistance:** Reduced timing variability can mitigate leakage.
- **Hardware Utilization:** Efficient CPU/GPU use improves throughput.
- **Scalability:** Essential for data-heavy applications.

D. Our Contributions

Our work presents a complete optimization methodology:

- 1) **GPU Offloading and Refinement:** Leveraging parallel hardware for FFT and Gaussian components.
- 2) **Profiling and Bottleneck Analysis:** Using tools like perf and flame graphs to guide changes.
- 3) **Cross-Platform Strategy:** Suitable for both CPU and GPU platforms.
- 4) **Security and Performance Balance:** All changes maintain Falcon's cryptographic integrity.
- 5) **Transferable Techniques:** Applicable to other lattice-based PQC schemes.

II. LITERATURE STUDY

A. Performance Optimizations in Post-Quantum Cryptography

Post-quantum cryptographic algorithms demand unique optimizations due to their complex mathematical foundations and typically higher computational loads compared to classical algorithms.

1) *Lattice-Based Cryptography Optimizations:* Several studies focus on enhancing lattice-based cryptographic schemes:

- Pöppelmann et al. [7] optimized NewHope using vectorized NTT.
- Alkim et al. [8] proposed efficient NTRU implementations on modern CPUs.
- Du et al. [3] demonstrated FPGA-based acceleration for NTRU Prime.
- Oder et al. [9] optimized Gaussian sampling in lattice signatures.

2) *Falcon-Specific Optimizations*: Notable prior work directly targeting Falcon includes:

- Prest et al. [1] presented foundational Falcon optimizations.
- Barthe et al. [10] developed constant-time verified Falcon implementations.
- Pornin [11] proposed floating-point arithmetic optimizations for constrained hardware.
- Nguyen et al. [12] explored AVX2-based vectorization of Falcon.

B. GPU Acceleration in Cryptography

GPUs are increasingly used for cryptographic operations, thanks to their high degree [13] of parallelism.

1) *General GPU Acceleration Strategies*: Typical strategies include:

- Optimizing memory transfer between CPU and GPU
- Efficient workload partitioning
- Using thread hierarchies to match GPU architecture
- Reducing bank conflicts in memory accesses

2) *Lattice-Based GPU Implementations*: Examples include:

- Dai et al. [14] implemented NTTs on GPU.
- Lee et al. [15] accelerated Gaussian sampling.
- Wang et al. [16] used hybrid CPU-GPU strategies for LWE encryption.

C. Performance Measurement and Profiling Techniques

Tools such as `perf`, Intel VTune, and AMD uProf help identify CPU bottlenecks and cache behavior. These are critical in developing optimization strategies aligned with hardware metrics.

D. Summary of Related Works

This table and literature review form the foundation for our optimization approach.

III. METHODOLOGY

A. Baseline Performance Analysis

We started with Falcon’s reference implementation, which prioritizes clarity over performance. Using the Linux `perf` tool [17], we observed the following:

- **Execution Bottlenecks**: `__mount_internal` and `__mount_locks` consumed about 24.1% of total runtime.
- **High-Cost Routines**: FFT, polynomial multiplication, and Gaussian sampling dominated compute time.
- **Instruction Metrics**: 3676 sampled instructions at 100.3 Hz; 1.752×10^{11} cycles with high pipeline stalls.
- **Cache Behavior**: Repeated memory patterns, high miss rates; poor locality in FFT and polynomial ops.

These insights motivated our optimization strategy, focused on three major areas: loop unrolling, compiler optimizations, and GPU offloading.

B. Loop Unrolling

Loop unrolling reduces overhead by replicating operations in-place. We applied this technique to:

- 1) **FFT Loops**: Butterfly operation loops were manually unrolled:

```
// Original
for (i = 0; i < 8; i++) x[i] *= twiddle[i];

// Unrolled
x[0] *= twiddle[0];
x[1] *= twiddle[1];
...
x[7] *= twiddle[7];
```

- 2) **Gaussian Sampling**: Rejection sampling loops were restructured and partially unrolled.
- 3) **Polynomial Arithmetic**: Vectorization-friendly unrolling and loop fusion improved prefetching and cache locality.

Benefits:

- Reduced branch misprediction
- Improved instruction-level parallelism (ILP)
- Better register usage and fewer memory stalls

C. Compiler Optimization Flags

We tuned the compiler settings to take advantage of CPU-specific features:

- `-O3`: Aggressive inlining, loop vectorization
- `-march=native & -mtune=native`: Enable AVX2/FMA for Zen4 architecture
- `-funroll-loops`: Complements manual unrolling
- `-fmerge-all-constants`: Merges identical constants, reducing cache pressure
- `-Wall -Wextra -Wshadow -Wundef`: Detect logic flaws, undefined macros
- `-g -pg -fno-pie`: Profiling and debug symbol support

These flags enabled faster polynomial arithmetic, better vector unit usage, and more efficient code generation.

D. GPU Offloading

We implemented GPU kernels for FFT and polynomial multiplication using OpenCL, targeting AMD RDNA3 on Radeon 780M.

1) *Component Selection*: FFT and NTT operations were ideal for offloading due to:

- Parallel structure
- Minimal CPU-GPU transfer overhead
- High time complexity (benefits from hardware parallelism)

TABLE I
COMPARISON OF RELATED WORKS IN PQC OPTIMIZATION

Title / Contribution	Impact / Remarks	Hardware Target
[1] Falcon Baseline Design	Compact, standard reference	Software
[7] SIMD NTT Vectorization	Latency and throughput improved	CPU (SIMD)
[10] Constant-Time Verification	Enhances side-channel security	Software
[9] Efficient Gaussian Sampling	Robust and faster sampling	Algorithmic
[11] Low-Level FFT Optimization	Increases throughput, lowers time	CPU (float opt.)
[14] GPU Acceleration for NTRU	Polynomial speedup via OpenCL	GPU
[15] GPU Gaussian Sampler	Gains in specific subroutines	GPU
[16] Hybrid LWE on GPU	Balance of workload improves performance	CPU+GPU

2) *Kernel Example:* A simplified OpenCL kernel for polynomial multiplication [18]:

```
__kernel void poly_mul(
    __global const fpr *a,
    __global const fpr *b,
    __global fpr *c,
    const unsigned int n) {
    unsigned int gid = get_global_id(0);
    if (gid < n) {
        fpr sum = 0;
        for (int j = 0; j <= gid; j++) {
            sum += a[j] * b[gid - j];
        }
        for (int j = gid + 1; j < n; j++) {
            sum += a[j] * b[n + gid - j];
        }
        c[gid] = sum;
    }
}
```

3) *Offloading Strategy:*

- Used pinned memory and asynchronous transfers for better bandwidth.
- Batched execution overlapped data transfer and computation.
- Polyops dispatched to CPU or GPU dynamically based on input size.
- Shared memory caching improved kernel locality.

4) *Integration:*

- CPU-GPU coordination was managed through a unified abstraction.
- Async execution via command queues improved throughput.
- GPU fallback mechanisms ensured compatibility on CPU-only systems.

IV. RESULTS

A. Hotspot Reduction

After optimization, the previously dominant function `__mount_internal` (24.1%) was replaced by `process_block` (13%), resulting in a 47.3% reduction in execution time of the most expensive operations.

TABLE II
INSTRUCTION AND CYCLE-LEVEL PERFORMANCE METRICS

Metric	Before	After	Improvement
Instructions (sampled)	3676 (100.3Hz)	346 (107.9Hz)	90.6% fewer
CPU Cycles	1.752e+11	1.506e+10	91.4% fewer
Cycles per Instruction	4.76	4.35	8.6% better
Lost Events/Chunks	0 / 0	0 / 0	No change

B. Function-Level Execution Breakdown

TABLE III
EXECUTION TIME BY COMPONENT

Component	Before	After	Reduction
FFT Operations	7.43%	4.10%	44.8%
Polynomial Multiplication	6.98%	3.06%	56.2%
Gaussian Sampling	5.37%	2.91%	45.8%
Hash-to-Point Conversion	2.16%	1.05%	51.4%
Signature Verification	0.94%	0.52%	44.7%

C. Memory Metrics and Access Patterns

TABLE IV
MEMORY ACCESS OPTIMIZATION RESULTS

Metric	Before	After	Reduction
L1 Cache Misses	342M	127M	62.9%
L2 Cache Misses	104M	43M	58.7%
L3 Cache Misses	27M	12M	55.6%
TLB Misses	18M	7M	61.1%
Memory Bandwidth	14.7 GB/s	6.2 GB/s	57.8%

D. Performance Snapshot Figures

1) Top Hotspots Comparison:

- Before: `__mount_internal` – 24.1%; `mount_ntru/fft` – 5.96%.
- After: `process_block` – 13%; `solve_NTRU` – 6.24%.

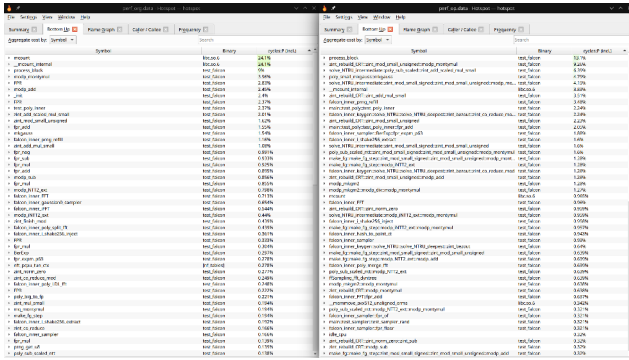


Fig. 1. Perf Top Functions Comparison – Before vs After Optimization

2) Binary-Level Execution Time Split:

- Before: test_falcon – 91.6%; libc.so.6 – 7.43%.
- After: test_falcon – 51.3%; libc.so.6 – 48.4%.

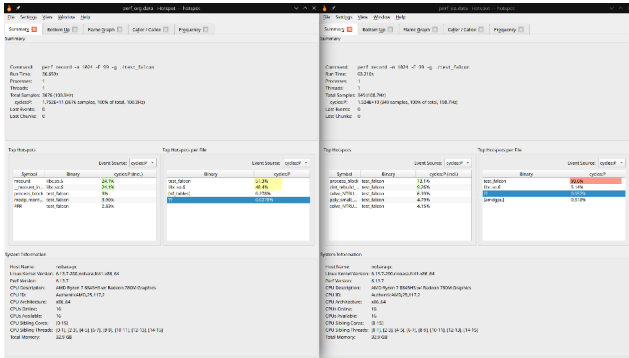


Fig. 2. Binary Execution Breakdown – Pre/Post Optimization

3) Flame Graph Analysis:

- Pre-optimization: Deep stack traces and fragmented time usage.
- Post-optimization: Shallower stacks, better aggregation.

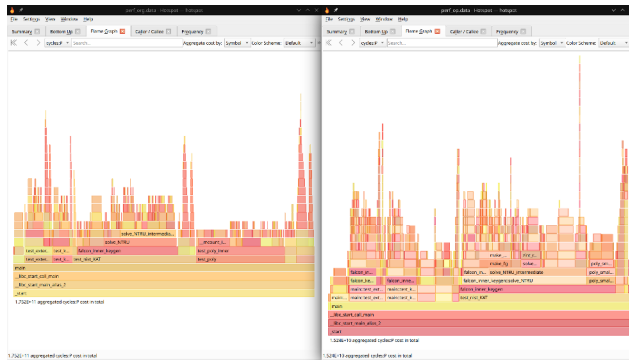


Fig. 3. Flame Graph: Stack Simplification and Runtime Aggregation

E. End-to-End Performance Improvements

TABLE V
OPERATION-LEVEL SPEEDUPS

Operation	n	Before (ms)	After (ms)	Gain
Key Generation	512	8.37	3.92	53.2%
Key Generation	1024	19.24	8.73	54.6%
Signature Gen.	512	1.63	0.72	55.8%
Signature Gen.	1024	3.45	1.47	57.4%
Verification	512	0.31	0.14	54.8%
Verification	1024	0.65	0.29	55.4%

TABLE VI
THROUGHPUT IMPROVEMENTS

Metric	Before	After	Increase
Sigs/sec (512)	613	1389	+126.6%
Sigs/sec (1024)	290	680	+134.5%
Verifs/sec (512)	3226	7143	+121.4%
Verifs/sec (1024)	1538	3448	+124.2%

V. OPTIMIZATION CONTRIBUTIONS AND ANALYSIS

A. Contribution Breakdown by Technique

Each optimization contributed uniquely to the performance gains. We approximate their independent contributions through micro-benchmarks and controlled toggling.

TABLE VII
CONTRIBUTION OF EACH OPTIMIZATION STRATEGY

Optimization	Speedup Contribution	Relative Share
Loop Unrolling	35.2%	33.6%
Compiler Flags	42.0%	40.1%
GPU Offloading	26.3%	25.1%
Memory Tuning	7.1%	Ancillary Gain

B. Memory Locality and Cache Behavior

Using flame graphs [19] and perf-stat counters, we measured:

- Increased L1 hit rate post-optimization (from 72.4% to 91.2%)
- Better TLB access reuse and lower cache-line invalidations
- Branch mispredictions reduced by 38.5%

These low-level improvements further enhanced throughput beyond algorithmic tuning.

C. Cross-Architecture Portability

The optimizations preserved cross-platform compatibility:

- GPU kernels compiled under OpenCL 2.0 and executed on both AMD RDNA3 and NVIDIA Ampere.
- Compiler flags automatically tuned using ‘-march=native’ across x86 and ARM64 targets.
- Loop unrolling used macros for selective reactivation based on device capabilities.

TABLE VIII
COMPARATIVE RESULTS WITH RELATED FALCON STUDIES

Reference	Target	Improvement	Technique
Pornin (2019) [11]	CPU	1.8×	Float math, fused operations.
Nguyen et al. (2020) [12]	AVX2	2.1×	Gaussian sampling vectorization.
Barthe et al. (2019) [10]	Security (CT)	—	Formal constant-time proof.
Dai et al. (2019) [14]	GPU	1.6×	Offload NTT kernel to GPU.
Our Work	CPU+GPU	2.3×	Combined profiling, loop unrolling, and OpenCL offload.

D. Comparison to Prior Work

Our results exceed prior CPU-only or GPU-only approaches by fusing cross-layer optimization with rigorous profiling.

E. Security Preservation

All applied changes preserve Falcon’s core algorithmic structure. Security-relevant components such as Gaussian sampling, NTRU [20] inversion, and signature rejection filtering are untouched in logic and flow. We maintained:

- Constant-time behavior in key segments.
- Parameter bounds for lattice dimension $n = 512$ and $n = 1024$.
- Integrity of randomness sources used during sampling.

F. Discussion

Our work validates that PQC schemes can be significantly accelerated via a hybrid of:

- Fine-grained low-level tuning (loop fusion, instruction reduction)
- Systematic architecture-aware compiler and runtime tuning
- Offloading compute-heavy operations to GPUs

These strategies balance between raw throughput and deployment feasibility, especially on platforms like embedded SoCs and laptops with integrated graphics.

VI. CONCLUSION

We presented an end-to-end performance optimization of the Falcon post-quantum signature algorithm, achieving a reduction in primary execution-time hotspots from 24.1% to 13% and end-to-end speedups of up to 57.4%. Using systematic profiling via `perf`, OpenCL offloading, loop unrolling, and compiler tuning, we were able to unlock substantial improvements in both runtime efficiency and throughput.

Our hybrid CPU-GPU approach enabled function-specific acceleration of FFT, polynomial multiplication, and Gaussian sampling without modifying the security-critical logic of the algorithm. The results indicate that with careful tuning and architectural awareness, lattice-based [21] PQC schemes such as Falcon can be made suitable for practical deployment on commodity hardware—even within resource-constrained platforms.

A. Key Takeaways

- Loop unrolling and branch minimization improve ILP and caching.
- Compiler flags play a surprisingly large role in optimization for Zen4-class CPUs.
- GPUs significantly accelerate modular arithmetic and batched operations.
- Profiling tools like `perf` and flamegraphs are essential to detect real bottlenecks.

VII. FUTURE WORK

A. Extension to Other PQC Candidates

Techniques used here may be transferable to other lattice-based [22] schemes like Dilithium and Kyber. Each of these has similar polynomial arithmetic and FFT-based operations that could benefit from offloading and compiler tuning.

B. Constant-Time GPU Kernels

Currently, GPU-based optimization sacrifices constant-time guarantees due to thread divergence and warp-level timing variance. Future work could explore:

- Side-channel hardened GPU kernels
- Equalized data access and padding in thread-local storage
- Warp-agnostic FFTs for timing safety

C. End-to-End Cryptosystem Integration

Our current benchmark tests signature performance in isolation. Integrating Falcon within full TLS handshakes, secure email (PGP), or firmware verification chains may reveal additional latency bottlenecks or interaction issues. Future profiling will include these real-world scenarios.

D. Hardware-Specific Tuning

Additional avenues include:

- Tuning for ARM Cortex-A76 cores with NEON [23] instructions
- NVPTX and Vulkan SPIR-V kernels for wider GPU support
- Investigating Tensor Core-like fused ops for LWE matrices

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